



"The learning companion "

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MODULE - 1

Basic concepts regarding electrical

- 1) charge - Electrical property of a substance
- 2) Current
- 3) Voltage
- 4) power
- 5) Energy

1) charge (Q)

Electrical property of a substance.

unit coulomb (C)

$$Q = It$$

2) Current (I)

It is the directed flow of free e^- s. Its direction is just opposite to the flow of e^- .

unit Ampere (A)

$$I = Q/t$$

3) Voltage / emf / potential (V)

workdone per charge (work/charge)

unit = J/C or Volt.

$$V = W/Q$$



pd b/n A & B

$$V_{AB} = V_A - V_B \\ = 5 - 3 = 2V$$

4) power (P)

Rate of doing work is called as power

unit = J/sec or watts

$$P = VI$$

5) Energy (E)

Capacity to do work

unit = J / wh (watt hour) / kwh

$$E = P \cdot t$$

1 unit = 1 kwh.

Circuit

It is a closed path. It is the interconnection of various circuit elements.

Circuit elements can be of 2 types

- 1) passive elements :- element that doesn't generate energy. Eg: Resistor, inductor & capacitor.
- 2) Active elements

↓
capable of generating energy

Eg: Battery, Generator etc.

RESISTOR

Resistance (R) :-

It is a property of a substance by which it opposes the flow of current

Unit = ohm (Ω)

Resistance of a material depends upon

- 1) length, $R \propto L$
- 2) Area, $R \propto 1/A$
- 3) Nature of the material
- 4) Temp.

$$R \propto L/A$$

$$R = \rho L/A$$

ρ = Resistivity / specific Resistance of the material

$$\rho = \frac{RA}{L} = \Omega m^2/m$$

$$\rho = \Omega m$$

Conductance (G) :-

It is the reciprocal of Resistance

$$G = 1/R$$

Ohm's Law

It states that the ratio of p.d (V) applied across the ends of the conductor to the current flowing through the conductor (I) is always a constant.

OR

Ohm's law states that the p.d applied across the ends of the a conductor is directly proportional to the current flowing through it.

$$\frac{V}{I} = \text{a const} \Rightarrow \frac{V}{I} = R$$

$$V \propto I$$

$$V = IR$$

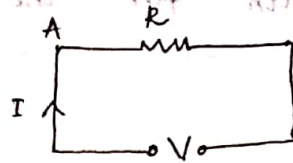
$P = VI$
$P = I^2 R$
$P = \frac{V^2}{R}$

$$P = IR \cdot I$$

$$= I^2 R$$

$$P = \frac{V \cdot V}{R}$$

$$= \frac{V^2}{R}$$



DC CIRCUIT

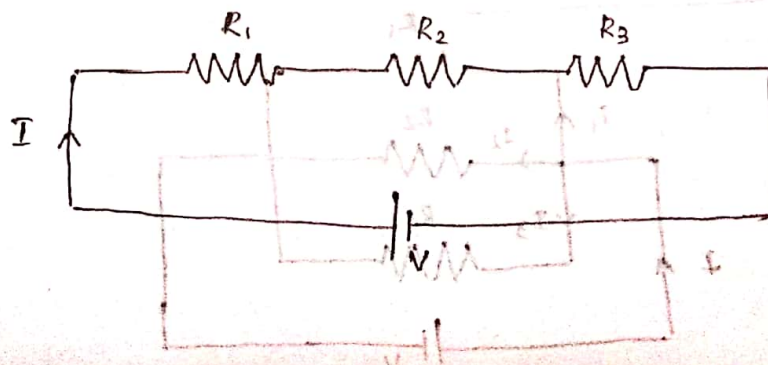
A circuit consist of the following components

- 1) Source of power
- 2) Conductor for carrying current
- 3) load.

There are 2 types of DC circuit

- 1) DC series circuit
- 2) DC parallel circuit

Dc Series Circuit



In a series circuit each resistance are connected end to end such that there is only one path for the current through the circuit.



Let V_1 , V_2 , & V_3 be the voltages across R_1 , R_2 & R_3 respectively. Then

$$V = V_1 + V_2 + V_3$$

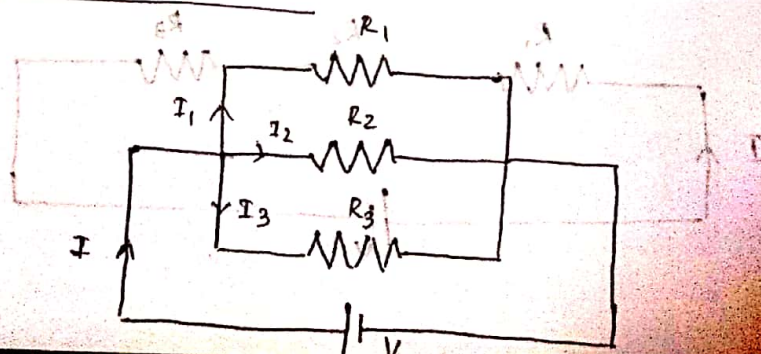
$$IR = IR_1 + IR_2 + IR_3$$

$$R_s = R_1 + R_2 + R_3$$

where R is the total or equivalent resistance of the series circuit.

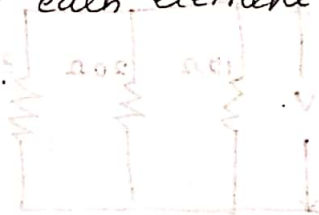
$$\text{Power } (P) = P_1 + P_2 + P_3$$

Dc parallel circuit



In parallel circuit one end of each resistor is connected to a common point & other end to another common point so, that current in the circuit will have different paths.

Voltage across each element is same in a parallel circuit.



$$\text{Total current } (I) = I_1 + I_2 + I_3$$

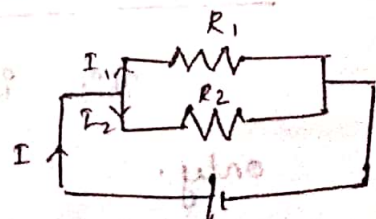
$$\frac{V}{R} = \frac{V}{R_1} + \frac{V}{R_2} + \frac{V}{R_3}$$

$$\boxed{\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}}$$

ie; Reciprocal of Total Resistance in a parallel circuit equal to the sum of the reciprocals of individual resistances.

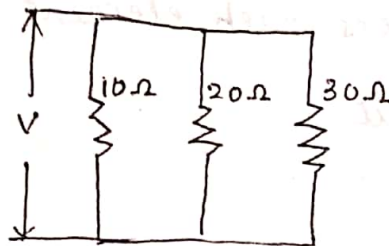
$$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2}$$

$$* \quad \boxed{R = \frac{R_1 R_2}{R_1 + R_2}}$$



$$P = P_1 + P_2 + P_3$$

* Find the equivalent resistance of the given network



→

$$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

$$= \frac{1}{10} + \frac{1}{20} + \frac{1}{30}$$

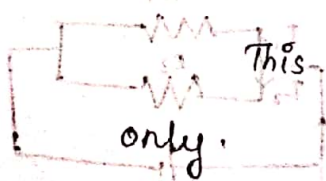
$$= 0.1 + 0.05 + 0.0333$$

$$\frac{1}{R} = \underline{0.1833}$$

$$\therefore R = \underline{5.4545 \Omega}$$

equivalent resistance is less than the smallest resistance in parallel circuit

VOLTAGE DIVISION RULE

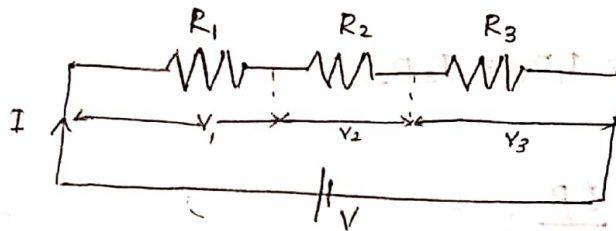


This is applicable in series circuits

only.

$$R = \frac{R_1 R_2}{R_1 + R_2}$$

if we have 2 resistances R_1 & R_2 connected in series and V be the total Voltage applied across the combinatⁿ. Let I be the current flowing through the circuit. According to Voltage divider rule



$$V_1 = IR_1$$

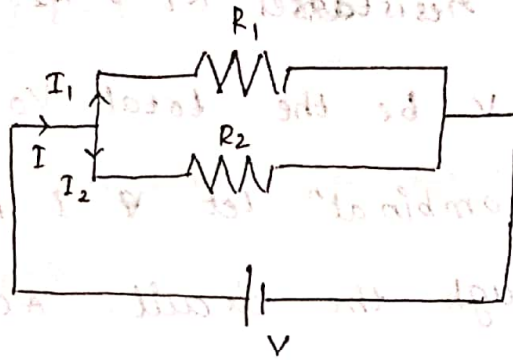
$$V_1 = \frac{V}{R} R_1$$

$$V_1 = \left(\frac{V}{R_1 + R_2} \right) R_1 = \frac{\text{Total Voltage}}{\text{Total Resist}} \times R_1$$

$$V_2 = \left(\frac{V}{R_1 + R_2} \right) R_2$$

CURRENT DIVISION RULE

it is appluable only in p^r circuit.



$$V = V_1 = V_2$$

$$IR = I_1 R_1 = I_2 R_2$$

$$I_1 = \frac{IR}{R_1}$$

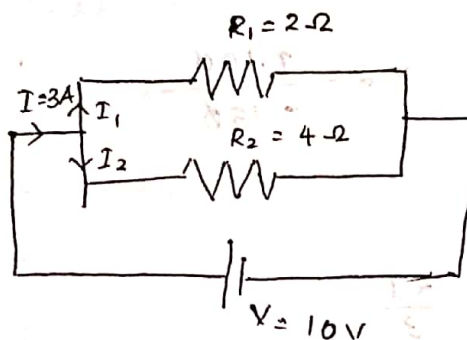
where R is the equivalent resistance of the parallel combination.

$$I_1 = \frac{I \frac{R_1 R_2}{R_1 + R_2}}{R_1} = \frac{I R_2}{R_1 + R_2}$$

$$I_1 = \frac{I R_2}{R_1 + R_2}$$

$$I_2 = \frac{I R_1}{R_1 + R_2}$$

*



$$I_1 = ?$$

$$I_2 = ?$$

$$\rightarrow I_1 = \frac{I R_2}{R_1 + R_2} = \frac{3 \times 4}{2 + 4} = \frac{12}{6} = \underline{\underline{2\text{A}}}$$

$$I_2 = \frac{I R_1}{R_1 + R_2} = \frac{3 \times 2}{2 + 4} = \frac{6}{6} = \underline{\underline{1\text{A}}}$$

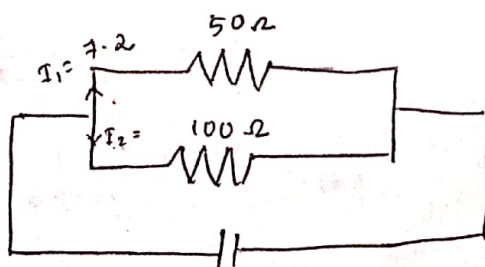
* A 50Ω resistor is in parallel with a 100Ω resistor.

Current in 50Ω is 7.2A . What is the value of

third resistance to be added in parallel to this

circuit to make total current 12A

→

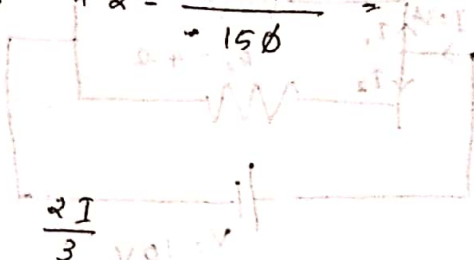


$$\frac{5000}{150} = 33.3$$

$$12 - 7.2 = 4.8$$

$$I_1 = \frac{IR_2}{R_1 + R_2}$$

$$7.2 = \frac{I \times 100}{150}$$



$$7.2 = \frac{10I}{153} = \frac{2I}{3}$$

$$\frac{21.6}{2} = I = \underline{\underline{10.8 \text{ A}}}$$

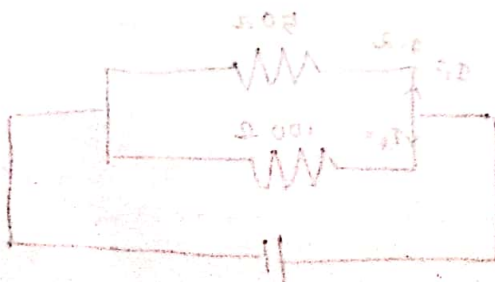
$$I_2 = \frac{IR_1}{R_1 + R_2} = \frac{10.8 \times 50}{150} = \frac{10.8}{3} = \underline{\underline{3.6 \text{ A}}}$$

$$I_3 = \frac{I R_{12}}{R_2 + R_3} = \frac{12.1 \times 150}{33.3} \quad \& \quad R = \frac{R_1 R_2}{R_1 + R_2} = 33.3$$

$$8.5 = \frac{1815}{33.3} \quad 402.93$$

$$150 + R_3 = \frac{402.93}{8.5} = 213.5$$

$$R_3 = 213.5 - 150 = 63.5$$



$$V = IR$$

$$V = V_1 = V_2 = V_3$$

$$V_1 = I_1 R_1$$

$$V = I R$$

$$R = \frac{V}{I}$$

$$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

$$IR = \frac{I R_1}{R_1 + R_2}$$

$$= \frac{12 \cdot 1 \times 50}{50 + 100} = 4.034 \text{ A}$$

$$I_1 = I_1 + I_2 + I_3$$

$$12.1 = 7.2 + 4.03 + I_3$$

$$= 0.87$$

$$I_1 = \frac{I R}{R_1} = 7.2 = \frac{12.1 R}{R_1}$$

$$R = \frac{7.2 \times 50}{12.1} = 29.75 \Omega$$

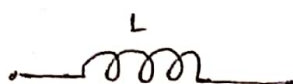
$$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

$$\frac{1}{R_3} = \frac{1}{29.75} - \frac{1}{50} - \frac{1}{100} = \frac{3.61}{1000} = 277 \Omega$$

INDUCTOR

is

A conductor, twisted to form a basic inductor. This is usually denoted by the letter L



(mag field produced by it)

when a current ' I ' flows through the inductor a mag field m will be set up around that inductor

$$\therefore \text{Inductance 'L'} = \frac{N\phi}{I} \Rightarrow \text{Henry (unit)}$$

N = no. of turns of the coil

I = Current

ϕ = mag flux in webber

when current through the inductor rises mag field expands & when the current decreases mag field will reduce.

ie; whenever there is a change in current, there will be a change in mag flux. According to Faraday's laws of

electromagnetic Induction,

An emf is (voltage) is induced in the inductor. i.e; Voltage across the inductor be proportional to $\frac{di}{dt}$

$$V \propto \frac{di}{dt}$$

$$V = L \frac{di}{dt} \quad \text{--- (1)}$$

$\Rightarrow$ power absorbed in the inductor

$$P = Vi$$
$$= L \frac{di}{dt} i$$

$$P = L i \frac{di}{dt} \quad \text{--- (2)}$$

$\Rightarrow$ Energy ~~also~~ (E)

$$E = P t$$

$$E = \int_0^t P dt$$

$$E = \int_0^{i^0} L i \frac{di}{dt} dt$$

$$E = \frac{L i^2}{2}$$

$$E = \frac{1}{2} L i^2$$

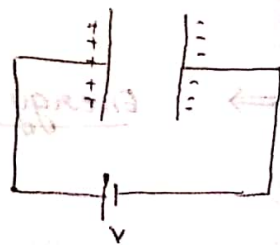
CAPACITOR

Two conducting plates are separated by a dielectric medium $\Rightarrow$ capacitor.
It is used to store electric charge.

Let Q be the charge stored on the capacitor & V be the potential applied across the capacitor

$$C = \frac{\text{charge}}{\text{voltage}} = \frac{Q}{V}$$

$$C = \frac{Q}{V} \Rightarrow \text{Faraday (unit)}$$



⇒ Current through the capacitor

$$I = \frac{dQ}{dt}$$

$$I = C \frac{dV}{dt}$$

$$C = \frac{Q}{V}$$

$$Q = CV$$

⇒ Power

$$P = VI$$

$$= \frac{Q}{C} V C \frac{dV}{dt}$$

$$P = CV \frac{dV}{dt}$$

⇒ Energy

$$E = Pt$$

$$= \int_0^t P dt = \int_0^t CV \frac{dV}{dt} dt$$

$$= \frac{CV^2}{2} = \frac{1}{2} CV^2$$

$$E = \frac{1}{2} CV^2$$

CONSTANT VOLTAGE SOURCE (Ideal Voltage Source)

It is defined as the energy source whose terminal voltage is independent of the current through it. It means that when the current drawn from the source varies from zero to infinity, the terminal voltage of the source remains unaffected. An Ideal Voltage source has zero internal resistance.



Const. dc
Voltage source

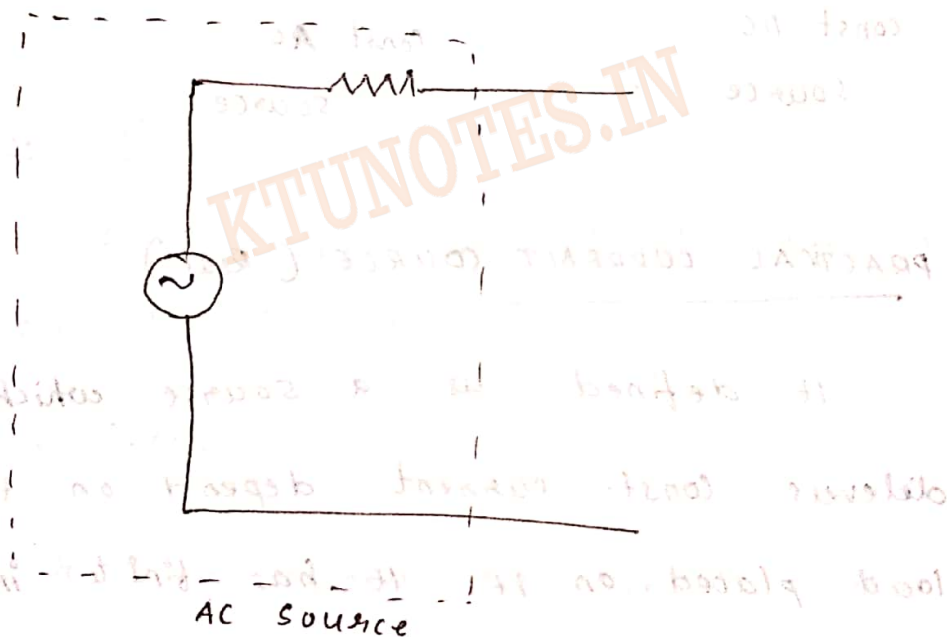
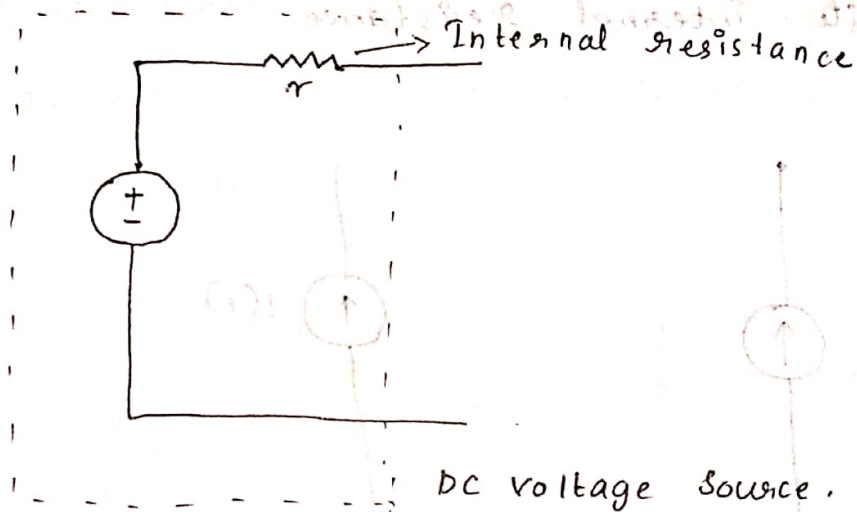


Const. AC
Voltage source

PRACTICAL VOLTAGE SOURCE (Real Voltage source)

It has low but finite internal resistance that causes its terminal voltage to decrease when the load current is increased.

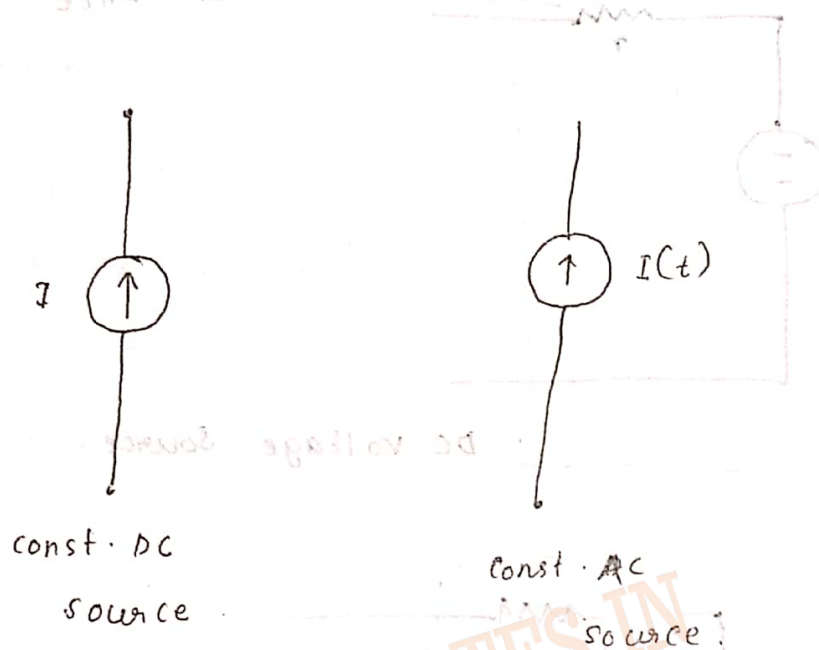
Vice versa. It can be represented by an ideal voltage source in series with a resistance.



CONSTANT (IDEAL) CURRENT SOURCE

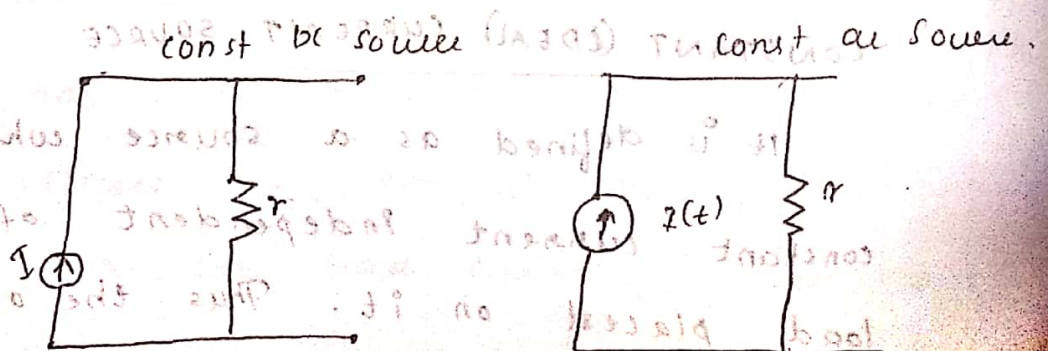
It is defined as a source which delivers constant current independent of the load placed on it. Thus the output

current of such a source remains unaltered from zero to infinity load. It has infinite internal resistance



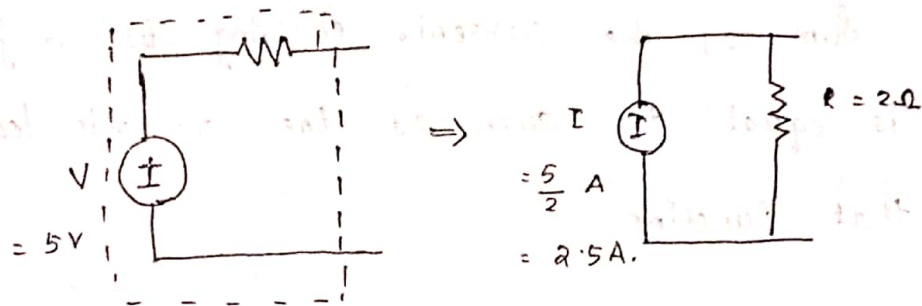
PRACTICAL CURRENT SOURCE (Real)

It is defined as a source which delivers const. current depend on the load placed on it. It has finite internal resistance



SOURCE CONVERSION

If we have a voltage source in series with a resistance can be converted into an equivalent current source in parallel with a resistor and vice versa.



NODE (Junction)

It is a point in a circuit where two or more elements meet.

LOOP (Mesh)

It is a closed path in a circuit.

KIRCHOFF'S LAWS

There are 2 laws.

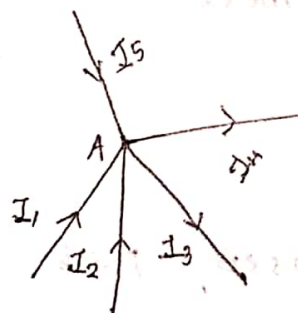
First law : Kirchhoff's Current Law (KCL) or Junction Rule.

It is associated with current in a circuit.
 The law states that the algebraic sum
 of the currents meeting at a point (node)
 is always zero.

OR

Sum of the currents entering to a junction
 is equal to sum of the currents leaving
 that junction.

The incoming currents can be taken as +ve &
 outgoing currents can be taken as -ve.



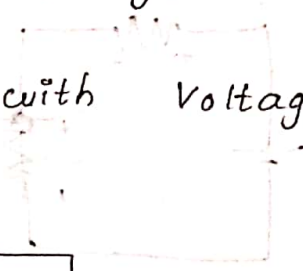
$$\Sigma I = 0$$

$$I_1 + I_2 - I_3 - I_4 + I_5 = 0$$

$$\underline{I_1 + I_2 + I_5 = I_3 + I_4}$$

IInd Law : Kirchoff's Voltage Law (KVL)

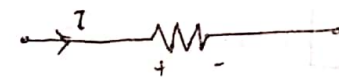
It is associated with voltages in a loop


$$\boxed{\sum IR + \sum \text{Emf} = 0}$$

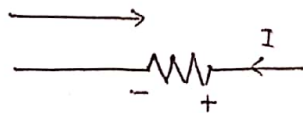
KVL states that algebraic sum of voltage drop across the resistors plus algebraic sum of emf in a closed mesh (loop) is always zero.

Sign Convention

①



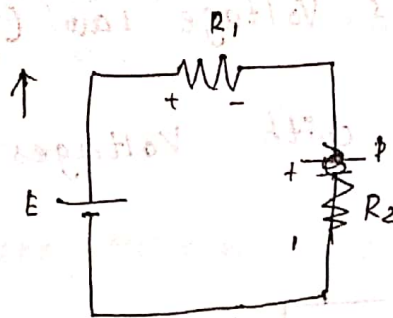
$+IR \Rightarrow$ Rise in potential.



②

$-IR \Rightarrow$ Fall in potential.

③.

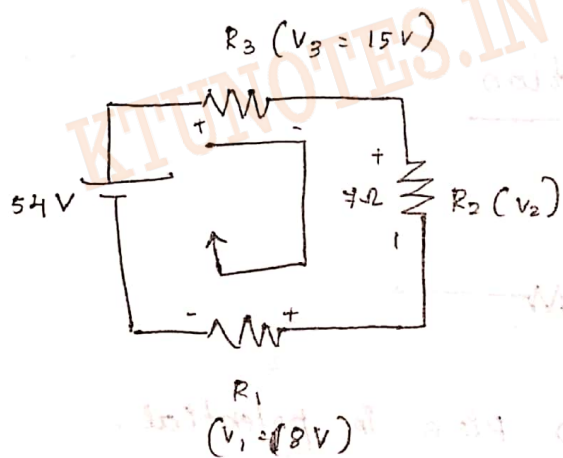


$$+IR_1 + IR_2 = 0 \quad E$$

$$IR_1 + IR_2 - E = 0$$

$$IR_1 + IR_2 = E$$

Q.



determine,

① V_2 using KVL

② current

③ R_1 & R_3

→

$$IR_3 + IR_2 + IR_1 - E = 0$$

$$IR_3 + IR_2 + IR_1 = E$$

$$I(R_1 + R_2 + R_3) = 54V$$

$I(1)$

$$18 + V_2 + 15 = 54V$$

$$V_2 = 54 - 15 - 18$$

$$= 21V$$

$$IR_2 = 21V$$

$$I \times 7 = 21V$$

$$I = 3A$$

$$IR_1 = 3 \times R_1$$

$$IR_1 = 18$$

$$R_1 = \frac{18}{3} = 6\Omega$$

$$IR_3 = 15$$

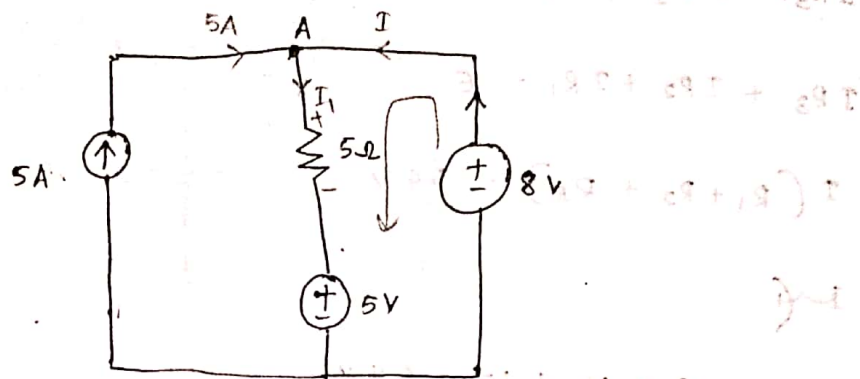
$$R_3 = \frac{15}{3} = 5\Omega$$



$$\frac{15+18}{33}$$

$$\frac{54-33}{21}$$

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Find I .

→ KCL at node A
2nd

$$+5A + I - I_1 = 0$$

$$5 + I = I_1 \quad \text{--- (1)}$$

KVL to 2nd mesh

$$5I_1 + 5 - 8 = 0$$

$$5I_1 = 3$$

$$I_1 = \frac{3}{5} \quad \text{--- (2)}$$

(2) in (1)

$$5 + I = \frac{3}{5}$$

$$\cancel{5} \frac{3}{\cancel{5}} = \frac{3}{5} - 5 = I$$

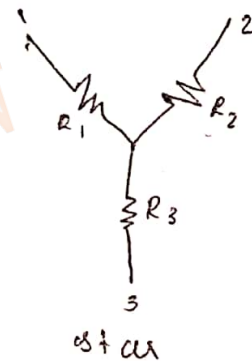
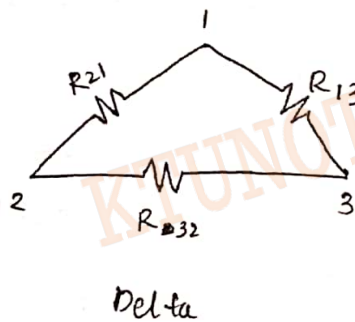
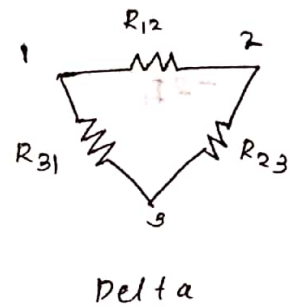
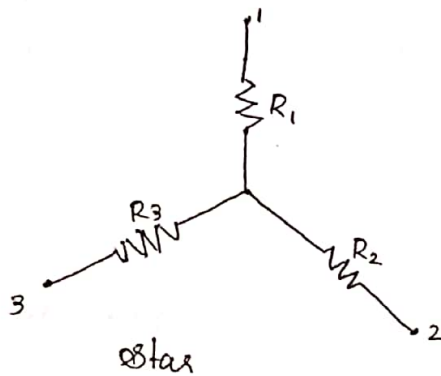
$$\frac{3 - 25}{5} = I$$

$$= -\frac{22}{5} = -4.4 \text{ A}$$

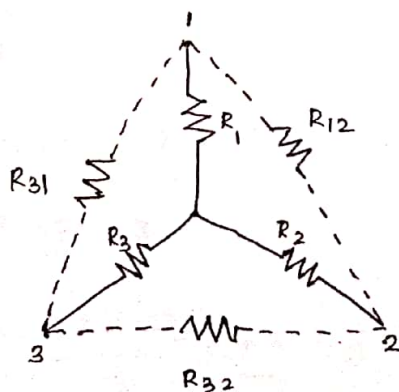
given directⁿ is opposite
to the actual direction.

Star to Delta & Delta to Star conversion

Y- Δ Δ -Y



Star to Delta Conversion



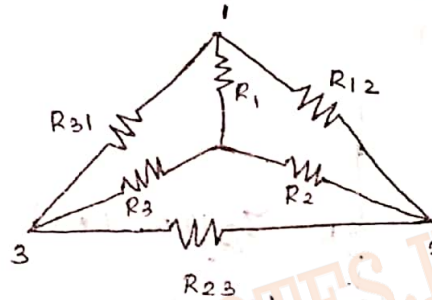
$$R_{12} = R_1 + R_2 + \frac{R_1 R_2}{R_3}$$

$$R_{32} = R_3 + R_2 + \frac{R_3 \cdot R_2}{R_1}$$

$$R_{31} = R_3 + R_1 + \frac{R_3 \cdot R_1}{R_2}$$



Delta to star conversion

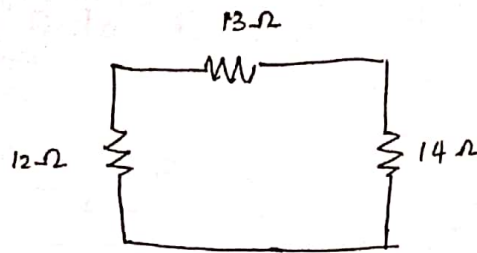


$$R_1 = \frac{R_{12} \cdot R_{31}}{R_{12} + R_{23} + R_{31}}$$

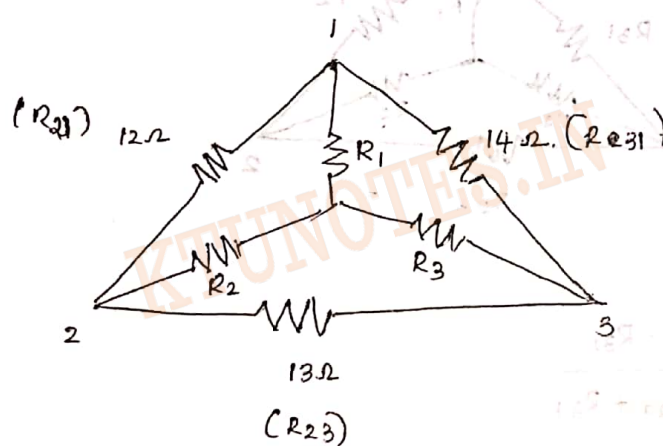
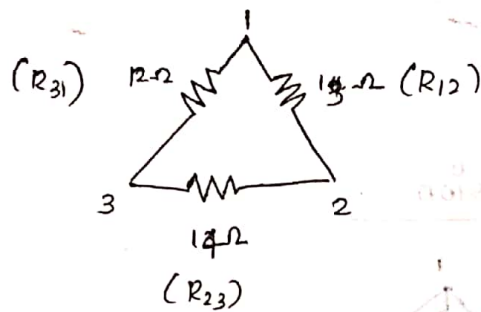
$$R_2 = \frac{R_{23} \cdot R_{12}}{R_{12} + R_{23} + R_{31}}$$

$$R_3 = \frac{R_{23} \cdot R_{31}}{R_{12} + R_{23} + R_{31}}$$

* obtain the star equivalent of a connected network as given.



→

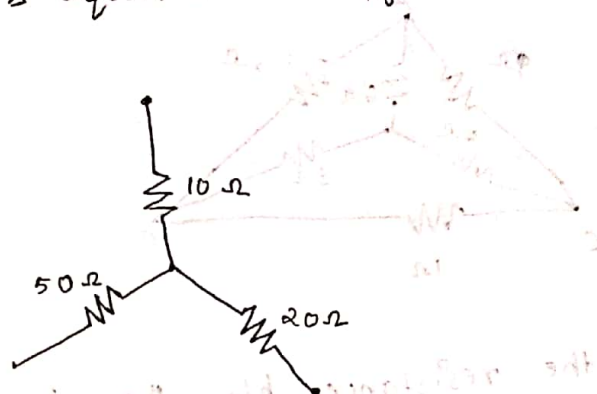


$$R_1 = \frac{R_{21} \cdot R_{31}}{R_{12} + R_{23} + R_{31}} = \frac{12 \times 14}{12 + 14 + 13} = \frac{168}{39} = 4.3$$

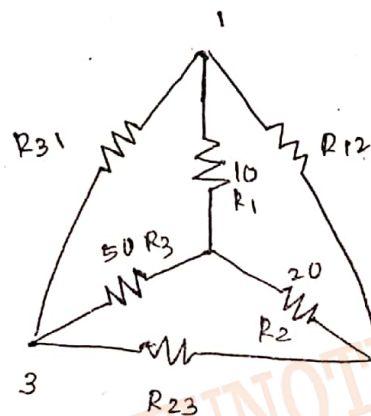
$$R_2 = \frac{R_{21} + R_{23}}{R_{12} + R_{23} + R_{31}} = \frac{12 \times 13}{12 + 14 + 13} = \frac{156}{39} = 4$$

$$R_3 = \frac{R_{31} + R_{23}}{R_{12} + R_{23} + R_{31}} = \frac{14 \times 13}{12 + 14 + 13} = \frac{182}{39} = 4.6$$

* obtain the Δ equivalent of the Y network



→



$$R_{31} = R_3 + R_1 + \frac{R_3 R_1}{R_2} = \frac{50 \times 10}{20} + 50 + 10 = \underline{\underline{85 \Omega}}$$

$$R_{12} = 10 + 20 + \frac{10 \times 20}{50} = \underline{\underline{34 \Omega}}$$

$$R_{23} = 20 + 50 + \frac{20 \times 50}{10} = \underline{\underline{170 \Omega}}$$

Q

Calculation

→



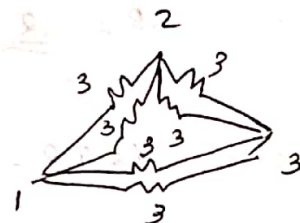
$$R_{12} = R_1 + R_2 + \frac{R_1 \cdot R_2}{R_3}$$

$$= 1 + 1 + \frac{1}{1}$$

$$= \underline{\underline{3}}$$

$$R_{32} = 3$$

$$R_{13} = 3$$



$$\frac{1}{R_{12}} = \frac{3}{3} + \frac{3}{3} = \frac{6}{9}$$

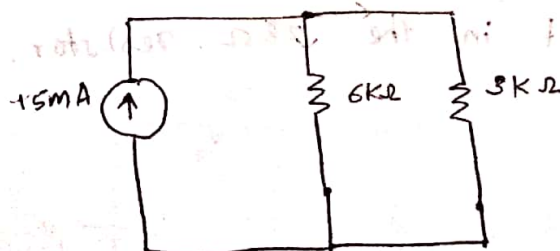
$$= \frac{2+1}{2} = \frac{3}{2} = \underline{\underline{1.5}}$$

~~R₁₂~~
 $R_{12} = \underline{\underline{0.67}}$

$$R_{12} = \frac{9}{6} = \frac{3}{2} = \underline{\underline{1.5}}$$

$$R_{AB} = \frac{1.5 \times 3}{1.5 + 3} = \underline{\underline{1}}$$

9)



Find current in 6kΩ resistor.

$$\frac{1}{R} = \frac{1}{6} + \frac{1}{3}$$

$$= \frac{3+6}{18} = \frac{9}{18} = \frac{1}{2} = \underline{\underline{0.5}}$$

$$R = \underline{\underline{2}}$$

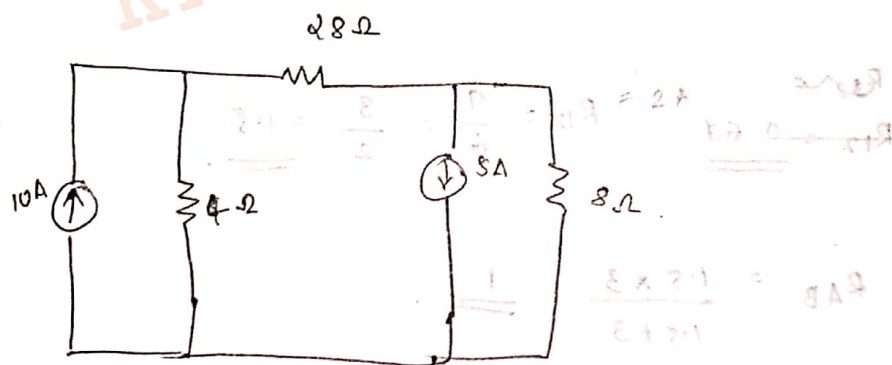
$$V = IR = 15 \times 10^{-3} \times 2$$

$$= \underline{\underline{30 \times 10^{-3}}}$$

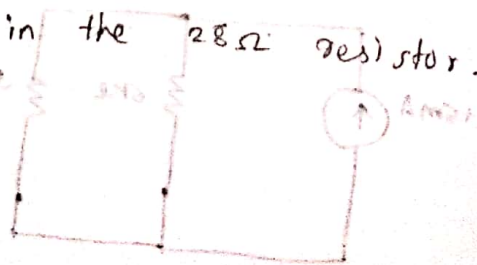
$$I = \frac{V}{R_1} = \frac{30 \times 10^{-3}}{6}$$

$$= \underline{\underline{5 \text{ mA}}}$$

Q)



Find the current in the 28Ω resistor.

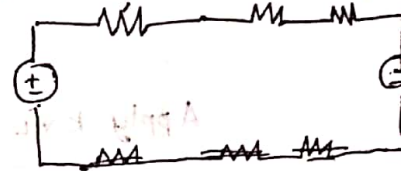


not clear and all the above hold

$$V_1 = \frac{I}{R} = \frac{105}{42} = \underline{\underline{\frac{5}{2}}}$$

$$V_1 = I_1 R_1 = 10 \times 4 = \underline{\underline{40V}}$$

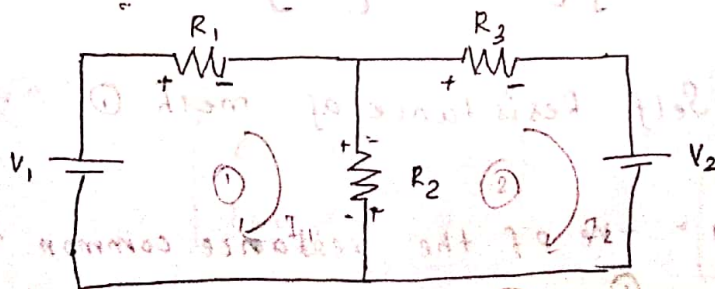
$$V_2 = I_2 R_2 = 5 \times 8 = 40V$$



MESH CURRENT METHOD (Maxwell's loop current method)

This method is based on KVL. 1st of In this method 1st of all we have to identify the meshes. Then current in each mesh is assigned in the clockwise direction.

Eg:



Apply KVL in mesh ①

$$\Sigma I_1 R_1 + I_1 R_2 - I_2 R_2 - V_1 = 0$$

$$I_1 (R_1 + R_2) - I_2 R_2 = V_1 \quad \text{--- (1)}$$



Apply KVL in mesh ②

$$I_2 R_2 + I_2 R_3 - I_1 R_2 + V_2 = 0$$

$$-I_1 R_2 + I_2 (R_2 + R_3) + V_2 = 0$$

$$-I_1 R_2 + I_2 (R_2 + R_3) = -V_2 \quad \text{--- (2)}$$

$$\begin{bmatrix} R_1 + R_2 & -R_2 \\ -R_2 & R_2 + R_3 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} +V_1 \\ -V_2 \end{bmatrix}$$

In general $[R][I] = [V]$

$$\begin{bmatrix} R_{11} & R_{12} \\ R_{21} & R_{22} \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} V_1 \\ V_2 \end{bmatrix}$$

R_{11} = Self Resistance of mesh ① (sum of all resistance in mesh ①)

$R_{12} = R_{21}$ = -ve of the resistance common to meshes ① & ②

Cal a -

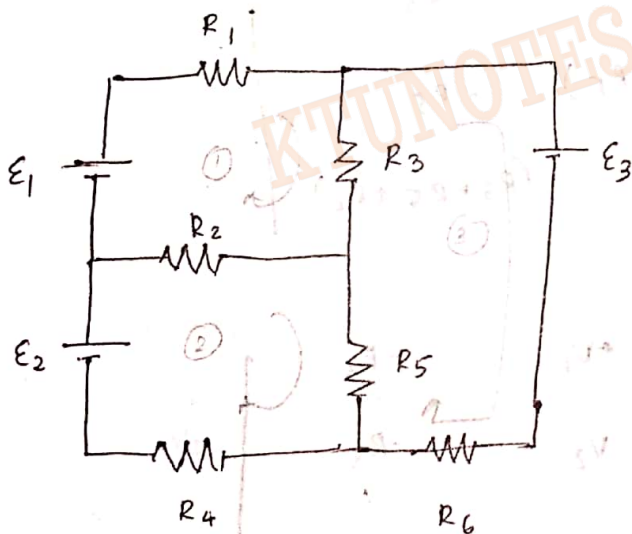
R_{22} = self-resistance of mesh ②

$$I_1 = \frac{\Delta_1}{\Delta} ; I_2 = \frac{\Delta_2}{\Delta}$$

$$\Delta = \begin{vmatrix} R_{11} & R_{12} \\ R_{21} & R_{22} \end{vmatrix}$$

$$\Delta_1 = \begin{vmatrix} V_1 & R_{12} \\ V_2 & R_{22} \end{vmatrix}$$

$$\Delta_2 = \begin{vmatrix} R_{11} & V_1 \\ R_{21} & V_2 \end{vmatrix}$$



→

$$[R][I] = [V]$$

$$[R] = \begin{bmatrix} R_{11} & R_{12} & R_{13} \\ R_{21} & R_{22} & R_{23} \\ R_{31} & R_{32} & R_{33} \end{bmatrix}$$

$$I = \begin{bmatrix} I_1 \\ I_2 \\ I_3 \end{bmatrix}$$

$$V = \begin{bmatrix} V_1 \\ V_2 \\ +V_3 \end{bmatrix}$$

$$\begin{bmatrix} (R_1+R_2+R_3) & -R_2 & -R_3 \\ -R_2 & (R_2+R_4+R_5) & -R_5 \\ -R_3 & -R_5 & (R_3+R_5+R_6) \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ I_3 \end{bmatrix} = \begin{bmatrix} V_1 \\ V_2 \\ -V_3 \end{bmatrix}$$

$$\Delta = \begin{vmatrix} (R_1+R_2+R_3) & -R_2 & -R_3 \\ -R_2 & (R_2+R_4+R_5) & -R_5 \\ -R_3 & -R_5 & (R_3+R_5+R_6) \end{vmatrix}$$

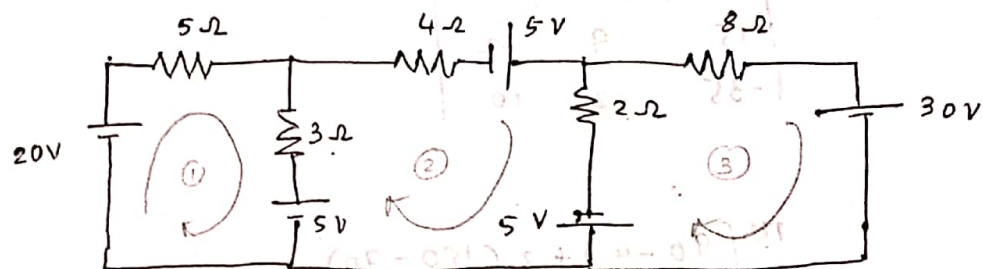
$$\Delta_1 = \begin{vmatrix} V_1 & -R_2 & -R_3 \\ V_2 & (R_2+R_4+R_5) & -R_5 \\ -V_3 & -R_5 & (R_3+R_5+R_6) \end{vmatrix}$$

$$\Delta_2 = \begin{vmatrix} R_1+R_2+R_3 & V_1 & -R_3 \\ -R_2 & V_2 & -R_5 \\ -R_3 & -V_3 & R_3+R_5+R_6 \end{vmatrix}$$

$$\Delta_3 = \begin{vmatrix} R_1+R_2+R_3 & -R_2 & V_1 \\ -R_2 & R_2+R_4+R_5 & V_2 \\ -R_3 & -R_5 & -V_3 \end{vmatrix}$$

Cal a-

* Find currents in each mesh using mesh current method.



→

$$[R][I] = [V]$$

$$R = \begin{bmatrix} R_{11} & R_{12} & R_{13} \\ R_{21} & R_{22} & R_{23} \\ R_{31} & R_{32} & R_{33} \end{bmatrix} = \begin{bmatrix} 5+3 & -3 & 0 \\ -3 & 3+4+2 & -2 \\ 0 & -2 & 2+8 \end{bmatrix}$$

$$I = \begin{bmatrix} I_1 \\ I_2 \\ I_3 \end{bmatrix} = \begin{bmatrix} I_1 \\ I_2 \\ I_3 \end{bmatrix}$$

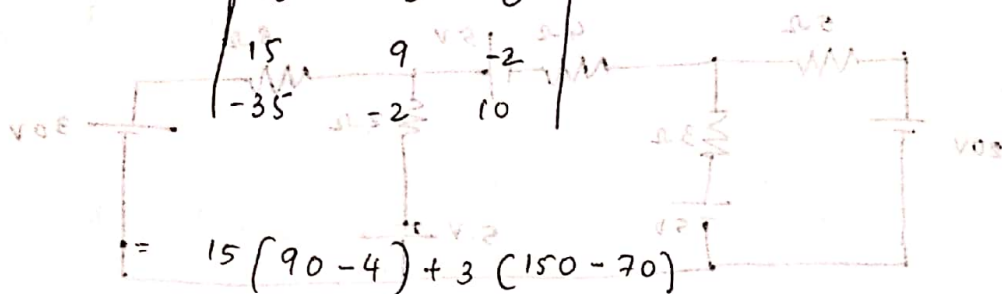
$$V = \begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix} = \begin{bmatrix} -5+20 \\ +5+5+5 \\ -30-5 \end{bmatrix}$$

$$\begin{bmatrix} 8 & -3 & 0 \\ -3 & 9 & -2 \\ 0 & -2 & 10 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ I_3 \end{bmatrix} = \begin{bmatrix} 15 \\ 15 \\ -35 \end{bmatrix}$$

$$\Delta = \begin{vmatrix} 8 & -3 & 0 \\ -3 & 9 & -2 \\ 0 & -2 & 10 \end{vmatrix} = 8(90-4) + 3(-30+0) = 8 \times 86 - 90 = 598$$

Find current in each branch of network by using loop method

$$\Delta_1 = \begin{vmatrix} 15 & -3 & 0 \\ 15 & 9 & -2 \\ -35 & -2 & 10 \end{vmatrix}$$



$$= 15(90 - 4) + 3(150 - 70)$$

$$= 15 \times 86 + 3 \times 80$$

$$= 1530$$

$$I_1 = \frac{1530}{598} = 2.5$$

$$I_2 = \frac{\Delta_2}{\Delta}$$

$$\Delta_2 = \begin{vmatrix} 8 & 15 & 0 \\ -3 & 9 & -2 \\ 0 & 2 & 10 \end{vmatrix}$$

$$= 8(90 + 4) - 15(30 - 0)$$

$$= 8 \times 94 + 450$$

$$= 1138$$

$$I_2 = \frac{1138}{598} = 1.9$$

$$\Delta_3 = \begin{vmatrix} 8 & -3 & 15 \\ -3 & 9 & 9 \\ 0 & -2 & 2 \end{vmatrix}$$

$$= 8(18 + 18) + 3(-6 + 0) + 15(6 + 0)$$

$$= -18 + 90$$

$$= 72$$

$$I_3 = \frac{72}{598} = 0.12$$

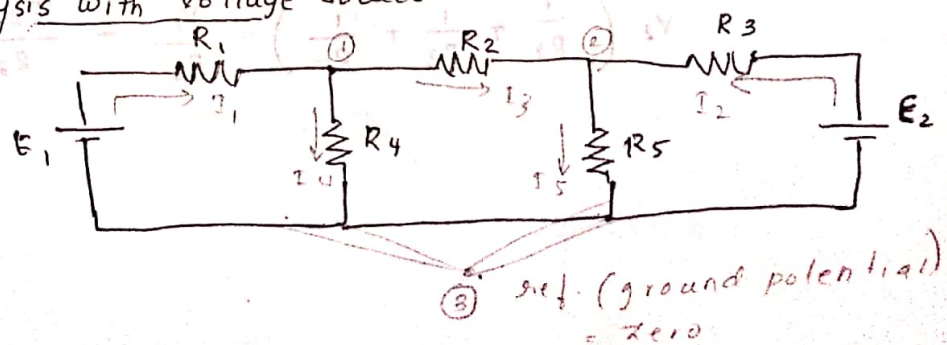
NODE VOLTAGE METHOD (Nodal Analysis)

This method is based on KCL

1. Identifying the nodes, one of the nodes can be taken as the reference node which is at ground potential.
2. if there are 'n' no. of nodes then we have to solve (n-1) equations.

Analysis with Voltage source

Eg:



$$n = 3$$

$$n-1 = 3-2 = 2 \text{ eqn are need to solve.}$$

$$\text{KCL at node ①} \Rightarrow I_1 - I_3 - I_4 = 0$$

$$\left(\frac{E_1 - V_1}{R_1} \right) - \left(\frac{V_1 - V_2}{R_2} \right) - \frac{V_1}{R_4} = 0$$

$$\frac{E_1}{R_1} - \frac{V_1}{R_1} - \frac{V_1}{R_2} + \frac{V_2}{R_2} - \frac{V_1}{R_4} = 0$$

$$-V_1 \left(\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_4} \right) + \frac{V_2}{R_2} = -\frac{E_1}{R_1}$$

$$V_1 \left(\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_4} \right) - \frac{V_2}{R_2} = \frac{E_1}{R_1} \quad \text{--- ①}$$

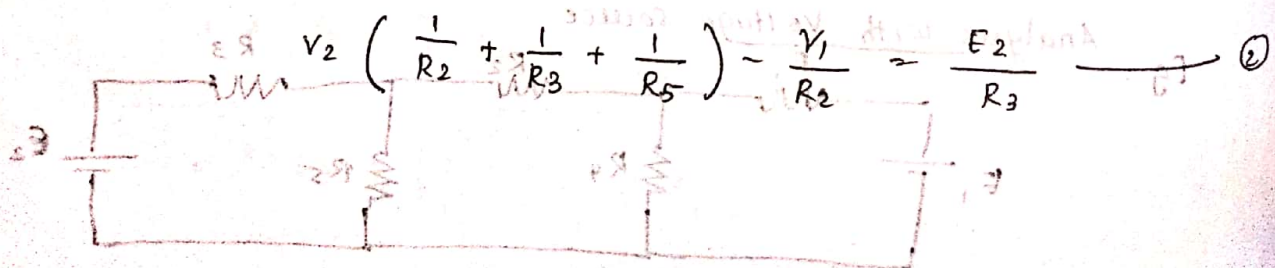
(Nodal Analysis) Node Voltage Method

$$\text{KCL at node ②} \Rightarrow I_3 + I_2 - I_5 = 0$$

$$\frac{V_1 - V_2}{R_2} + \frac{E_2 - V_2}{R_3} - \frac{V_2}{R_5} = 0$$

$$\frac{V_1}{R_2} - \frac{V_2}{R_2} + \frac{E_2}{R_3} - \frac{V_2}{R_3} - \frac{V_2}{R_5} = 0$$

$$-V_2 \left(\frac{1}{R_2} + \frac{1}{R_3} + \frac{1}{R_5} \right) + \frac{V_1}{R_2} = -\frac{E_2}{R_3}$$



Calculation

From ① & ② $\Rightarrow$

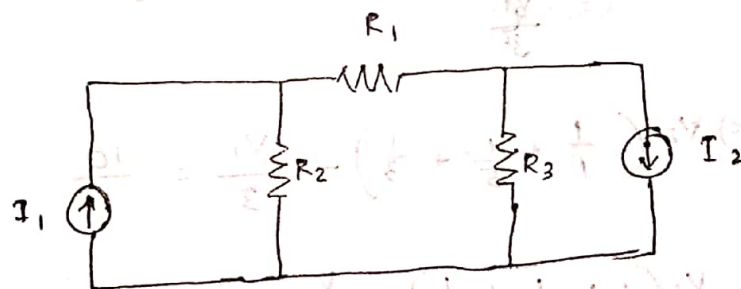
$$\begin{bmatrix} \left(\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_4}\right) & -\frac{1}{R_2} \\ -\frac{1}{R_2} & \left(\frac{1}{R_2} + \frac{1}{R_3} + \frac{1}{R_5}\right) \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} E_1/R_1 \\ E_2/R_2 \end{bmatrix}$$

$$V_1 = \Delta^{-1} \Delta^1 / \Delta$$

$$V_2 = \Delta^{-1} \Delta^2 / \Delta$$

$$\Delta = \begin{vmatrix} \left(\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_4}\right) & -\frac{1}{R_2} \\ -\frac{1}{R_2} & \left(\frac{1}{R_2} + \frac{1}{R_3} + \frac{1}{R_5}\right) \end{vmatrix}$$

Analysis with current sources

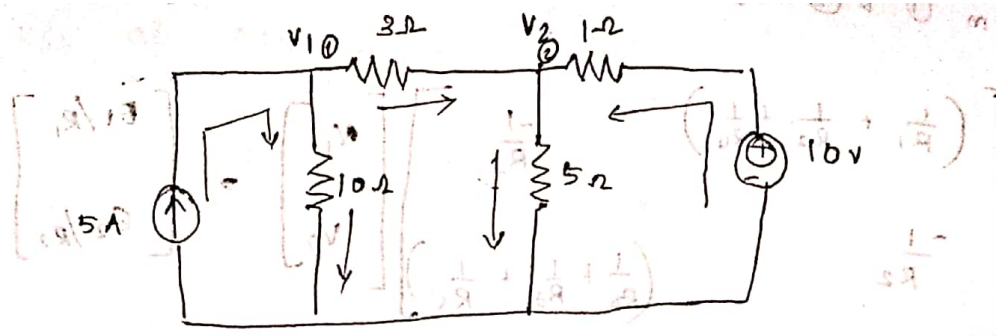


$$\text{Node ①} \Rightarrow V_1 \left(\frac{1}{R_1} + \frac{1}{R_2} \right) - \frac{V_2}{R_1} = I_1$$

$$\text{Node ②} \Rightarrow V_2 \left(\frac{1}{R_1} + \frac{1}{R_3} \right) - \frac{V_1}{R_1} = I_2$$

$$\begin{bmatrix} 2 \\ 0 \end{bmatrix} = \begin{bmatrix} 1V \\ 5V \end{bmatrix} \begin{bmatrix} \frac{1}{R_1} + \frac{1}{R_2} & -\frac{1}{R_1} \\ -\frac{1}{R_1} & \frac{1}{R_1} + \frac{1}{R_3} \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix}$$

Q7)



Write the node voltage eq^{ns} and determine the current in each branch for the network shown.

→ Node ① ⇒ $V_1 \left(\frac{1}{R_1} + \frac{1}{R_2} \right) - \frac{V_2}{R_3} = I_1$

$$V_1 \left(\frac{1}{3} + \frac{1}{10} \right) - \frac{V_2}{5} = 5$$

Node ② ⇒ $V_2 \left(\frac{1}{1} + \frac{1}{5} + \frac{1}{3} \right) - \frac{V_1}{5} = \frac{10}{1}$

$$= V_2 \left(1 + \frac{1}{5} + \frac{1}{3} \right) - \frac{V_1}{5} = 10$$

$$= V_2 \frac{23}{15} - \frac{V_1}{5} = 10$$

$$\begin{bmatrix} \frac{13}{30} & -\frac{1}{5} \\ -\frac{1}{5} & \frac{23}{15} \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} 5 \\ 10 \end{bmatrix}$$

$$\frac{23}{15} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \frac{83}{150} \begin{bmatrix} 5 \\ 10 \end{bmatrix}$$

$$D = 0.05633 = \frac{83}{150}$$

$$\Delta_1 = \begin{vmatrix} 5 & -V_3 \\ 10 & 23/15 \end{vmatrix} = 11$$

$$\Delta_2 = \begin{vmatrix} 13/30 & 5 \\ -V_3 & 10 \end{vmatrix} = 6$$

$$V_1 = \frac{\Delta_1}{D} = \frac{11}{83/150} = \frac{11}{0.55} = \underline{\underline{19.87V}}$$

$$V_2 = \frac{6}{0.55} = \underline{\underline{10.9V}}$$

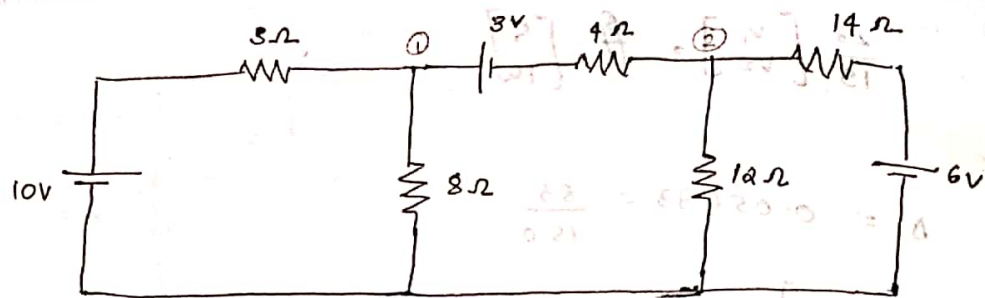
$$I_1 = \frac{V_1}{10} = \frac{19.87}{10} = \underline{\underline{1.987A}}$$

$$I_2 = \frac{V_1 - V_2}{3} = \frac{19.87 - 10.84}{3} = \underline{\underline{3.01A}}$$

$$I_3 = \frac{V_2 - 0}{5} = \frac{10.84}{5} = \underline{\underline{2.168A}}$$

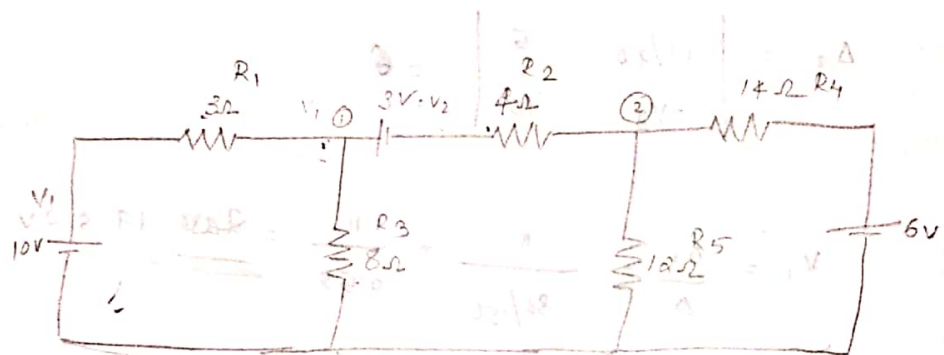
$$I_4 = \frac{10 - V_2}{10} = \frac{10 - 10.84}{10} = \underline{\underline{-0.084A}}$$

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 5 \\ 10 \end{bmatrix}$$



Find current through 8Ω & 12Ω resistors.

→



$$n = 3$$

$$n - 1 = 3 - 2 = 2 \text{ eqn}$$

$$\text{Node ①} \Rightarrow V_1 \left(\frac{1}{3} + \frac{1}{8} + \frac{1}{4} \right) - \frac{V_2}{4} = \frac{10}{3}$$

$$\text{Node ②} \Rightarrow V_2 \left(\frac{1}{4} + \frac{1}{12} + \frac{1}{14} \right) - \frac{V_1}{4} = \frac{6}{12}$$

$$\begin{bmatrix} \frac{1}{3} + \frac{1}{8} + \frac{1}{4} & -\frac{1}{4} \\ -\frac{1}{4} & \frac{1}{4} + \frac{1}{12} + \frac{1}{14} \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} \frac{10}{3} \\ \frac{6}{12} \end{bmatrix}$$

$$\begin{bmatrix} 0.708 & -0.25 \\ -0.25 & 0.404 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} 3.33 + \frac{3}{4} \\ 0.42 - \frac{3}{4} \end{bmatrix}$$

$$\Delta = \begin{vmatrix} 0.708 & -0.25 \\ -0.25 & 0.404 \end{vmatrix} = \begin{vmatrix} 17/24 & -1/4 \\ -1/4 & 17/42 \end{vmatrix}$$

$$= \frac{47}{336} = 0.224$$

$$\Delta_1 = \begin{vmatrix} 3.33 + \frac{3}{4} & -1/4 \\ 0.42 - \frac{3}{4} & 17/42 \end{vmatrix}$$

$$= 1.57$$

$$(4.08 \times 0.404) - (-0.25 \times 0.3) = 1.5848 = 0.09952$$

$$\Delta_2 = \begin{vmatrix} 17/24 & 10/14 + \frac{3}{4} \\ -1/4 & 6/14 - \frac{3}{4} \end{vmatrix}$$

$$= \frac{1585}{1668}$$

$$= 1.57 \frac{533}{672} = 0.7$$

$$V_1 = \frac{\Delta_1}{\Delta} = \frac{0.793}{\frac{47}{336}} = 5.67$$

$$V_2 = \frac{\Delta_2}{\Delta} = \frac{1.57}{0.139} = 11.22$$

$$I_1 = V_1$$

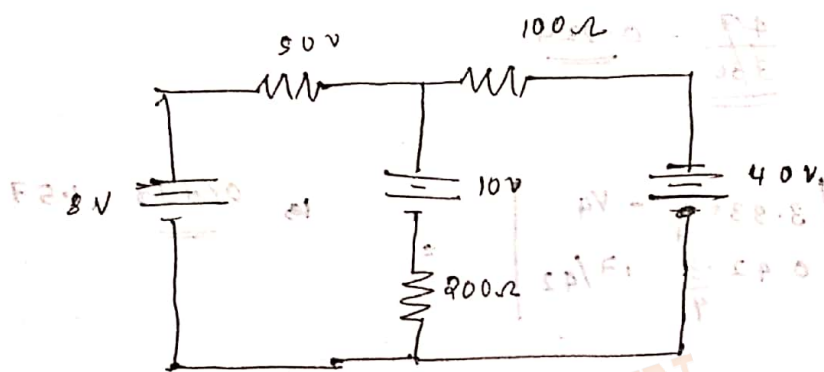
$$V_1 = \frac{1.57}{0.139} = 7.01$$

$$V_2 = \frac{0.793}{0.124} = 3.5 \text{ A V}$$

$$I_1 = \frac{V_1}{R} = \frac{2.01}{2} = 0.876 \text{ A}$$

$$I_2 = \frac{V_2}{R} = \frac{3.54}{12} = 0.2948 \text{ A}$$

Q Find power in each resistor



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$$P = VI = \frac{V^2}{R}$$

$$P = \frac{V^2}{R}$$

$$P = \frac{V^2}{R}$$

$$P = \frac{V^2}{R}$$

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